



Encephalitis

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Brain inflammation secondary to encephalitis is an urgent global emergency and presents multiple opportunities to reduce current substantial morbidity and mortality. Aetiologies can be divided into infectious and autoimmune causes. In this Seminar, we highlight pragmatic clinical approaches to recognise and distinguish the most common pathogenic viruses and emerging range of autoantibodies encountered in routine practice. These pre-test impressions are judiciously shaped by valuable, simple investigations—particularly serum and cerebrospinal fluid nucleic acid and autoantibody testing—to identify the precise causative agent. This clinically led approach ensures early recognition of encephalitis subtypes, facilitates the timely administration of antivirals and immunotherapies proven to improve patient outcomes, and minimises the frequent misdiagnosis of autoimmune encephalitis. Finally, we review emerging targeted therapeutic approaches, measurements of clinical encephalitis outcomes, and environmental and vaccine-centred strategies to improve prevention, diagnosis, and care for patients with encephalitis, cognisant of long-term patient, caregiver, and economic burdens.

Introduction

Encephalitis has long been one of medicine's most challenging and enigmatic syndromes, spanning the still-unexplained 20th century epidemic of encephalitis lethargica, the rapidly expanding spectrum of causative autoantibodies identified in the past two decades, and contemporary outbreaks driven by emerging viral pathogens. Characterised as brain inflammation, the umbrella of encephalitis represents a heterogeneous group of disorders with an increasingly recognised number of infectious and autoimmune causes. Despite the possibility of prevention and, in some cases, potential for reversibility with early therapeutic interventions, encephalitis has a high mortality rate with myriad sequelae, including life-changing disabilities.^{1,2} Encephalitis affects men and women of all ages and ethnicities worldwide, although epidemiological and more precise clinical features vary by cause. Global incidence data estimate 500 000–1.5 million people develop encephalitis each year.^{2–4} In 2021, encephalitis was the fourth leading cause of neurological health loss in children younger than 5 years, and 13th across all age

groups, with an estimated 100 000 people dying annually from the condition.^{1,3} Encephalitis imposes a substantial economic burden: hospitalisation estimates were approximately US\$2 billion in the USA⁵ and more than £23 million per year in England,⁶ excluding costs associated with intensive care, rehabilitation, and long-term dependence.

Moreover, WHO describes encephalitis as an “increasing global threat” with “an urgent public health imperative”.^{7–9} Recommended methods to avert this potential crisis include prevention, improved recognition, and timely treatment. Several forms of infectious encephalitis are preventable, with improved vaccine confidence and vaccination programmes being key to reducing incidence, death, and disability.¹⁰ Furthermore, early diagnosis and timely treatments are interdependent, and equally pivotal in improving patient outcomes and mortality. In herpes simplex virus (HSV) encephalitis, timely acyclovir administration reduces mortality from approximately 70% to around 20% ($p=0.03$).^{11,12} Similarly, forms of autoimmune encephalitis show the most impressive clinical improvements after early immunotherapy administration, with some studies observing poorer outcomes for each day of delayed immunotherapy.^{13–15} Prompt treatment administration in infectious encephalitis and autoimmune encephalitis is primarily facilitated by educating physicians to recognise the key presenting clinical features. Improving accurate recognition of encephalitis aims to avoid two well recognised problems: missed opportunities for earlier therapies and over-diagnosis. The latter is principally secondary to the misattribution of autoimmune encephalitis in the context of clinically irrelevant autoantibody results and multiple varied encephalopathic syndromes (figures 1–3).^{16,17}

Over the past two decades, incidence of infectious encephalitis has remained stable, primarily due to vaccinations and improved recognition. By contrast, diagnosis of autoimmune encephalitis has risen dramatically, driven by the discovery of pathogenic autoantibodies that have identified syndromes previously considered to be idiopathic.^{18–20} Overall, all-cause

Search strategy and selection criteria

To capture key recent publications in the field, Embase and MEDLINE databases were searched for articles published from Jan 1, 2022, to June 4, 2025, using the following terms: encephalitis, anti-N-methyl-D-aspartate receptor encephalitis, viral encephalitis, Japanese encephalitis, West Nile Fever, varicella zoster, and limbic encephalitis (appendix p 1). Only articles published in English, human studies, clinical studies or trials, observational studies, guidelines, multicentre studies, and practice guidelines were considered. 184 and 350 unique references were generated from the two databases, respectively. We also searched selected articles for further references that we judged relevant, in addition to landmark publications familiar to the authors. Review articles are cited as referenced summaries for topics outside the scope of this Seminar.

See Online for appendix

autoimmune encephalitis is at least as common as infectious encephalitis in high-income countries,^{21,22} with similar observations emerging in low-to-middle-income countries (LMICs).^{23,24} Autoimmune and infectious encephalitis, and their subtypes, should therefore be considered in parallel for the contemporary management of acute encephalitis.

In this Seminar, we use clinical observations to facilitate the accurate recognition of common forms of autoimmune and infectious encephalitis, and highlight established and emerging diagnostics. We also outline acute therapeutic strategies alongside promising avenues for precision medicine, and discuss how these approaches shape patient outcomes after encephalitis. When distinctive, we provide brief perspectives on encephalitis in LMICs, children, and immunocompromised hosts. Finally, we review recent developments and highlight outstanding research questions in encephalitis care, spanning prevention, impacts of climate change, clinical classifications, and enhanced molecular diagnostics.

What is encephalitis?

Within general medicine, neurology, paediatrics, psychiatry, and emergency medicine practices, regularly asked questions include: “is this encephalitis?” and, if so, “which form of encephalitis?” These questions reflect the importance of recognising syndromes that are potentially treatable or even reversible. Initially, answers can be guided by understanding core features that broadly distinguish encephalopathy from encephalitis (appendix p 2).²⁵ Encephalopathy typically manifests in patients over a few days with diffuse brain dysfunction, often with coma or disorientation (or both), but without focal neurological features consistent with localised disease. Generally, an underlying cause is identified, such as

toxins or metabolic disturbances (eg, liver or renal failure, or Wernicke–Korsakoff syndrome), systemic infection, hypertension, or cerebral hypoxia.^{26–28} Typically, MRI and routine cerebrospinal fluid (CSF) investigations appear unremarkable, whereas electroencephalography (EEG) shows diffuse slowing of cerebral activity, consistent with a global brain disturbance. By contrast, many forms of encephalitis (figure 1), particularly infectious encephalitis, present over a similar time course but with combinations of fever, new-onset seizures, focal deficits (eg, aphasia, hemiparesis, or brainstem signs), CSF pleocytosis, and, often, MRI hyperintensities. In addition to the causes of encephalopathies,^{26,28} other mimics of encephalitis include the myriad differential diagnoses within the category of rapidly progressive dementia, including treatable conditions such as cerebral vasculitis, inflammatory cerebral amyloid angiopathy, non-convulsive seizures, and new onset refractory status epilepticus.^{29,30} Distinguishing these conditions from forms of infectious and autoimmune encephalitis enables accurate diagnosis and helps guide specific treatments.

A clinical approach to differentiating common causes of encephalitis

Once encephalitis is considered likely (also see the Diagnostic investigations section and figure 4), management options differ depending on the subtype (ie, infectious vs autoimmune encephalitis). Broadly, these subtypes share the core features of seizures, and multiple cognitive and psychiatric disturbances. The diagnostic process for acute encephalitis should therefore involve parallel testing for causes of both subtypes to ensure timely diagnosis (figures 1–3). However, the nature and

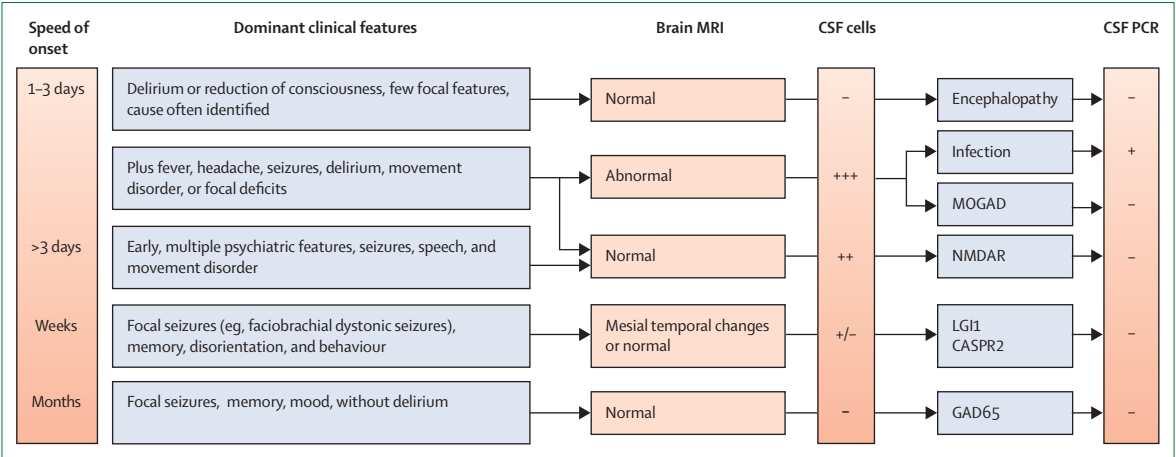
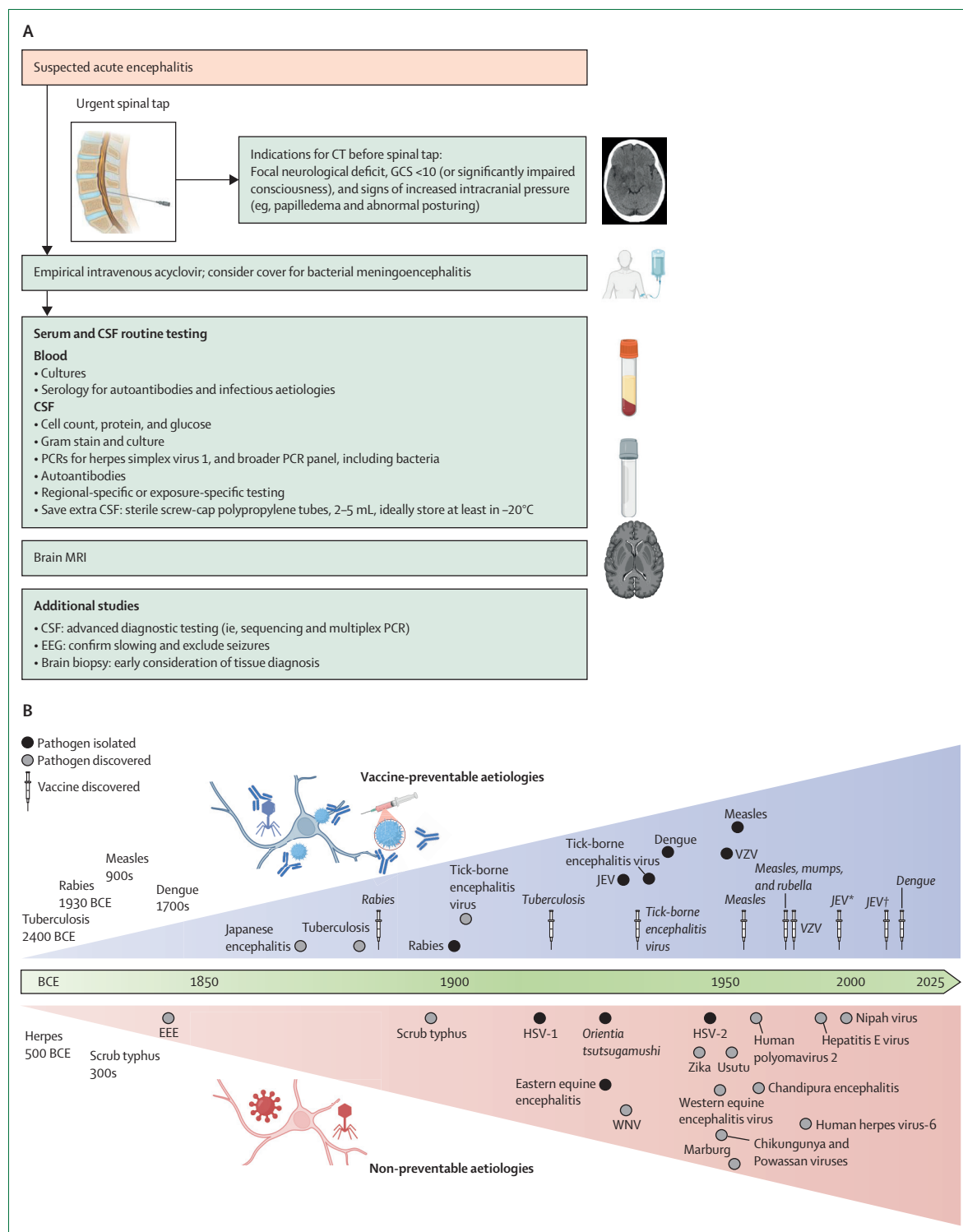


Figure 1: Pragmatic approach to distinguish common forms of encephalitis
Encephalopathy, infectious encephalitis, and forms of autoimmune encephalitis can be largely differentiated based on the speed of symptom onset, dominant clinical features, and by incorporating MRI, the presence of CSF cells, and PCR results. This clinical guide is intended to provide a simple, digestible approach to capture common encephalitis syndromes, without being definitive, and should be refined with data from figures 2 and 3, and appendix pp 4–29. CSF=cerebrospinal fluid. GAD65=glutamic acid decarboxylase 65. LG1=leucine-rich glioma-inactivated 1. MOGAD=myelin oligodendrocyte glycoprotein antibody-associated disease. NMDAR=N-methyl-D-aspartate receptor.

combinations of these clinical features and their onset duration can aid in distinguishing between infectious and autoimmune encephalitis, and, in many causes, their subtypes (figure 1).

Most forms of infectious encephalitis present with neurological symptoms a few days after a prodrome of systemic symptoms, with combinations of fever, headache, confusion, seizures, psychiatric changes, and



(Figure 2 continues on next page)

focal deficits. More detailed clinical observations and investigations can differentiate between infectious encephalitis aetiologies to guide appropriate management (figure 2B–C; appendix pp 4–16). Worldwide, the most common subtype of infectious encephalitis is mediated by herpes simplex virus (HSV)-1, with a bimodal age distribution mostly affecting people younger than 30 years, or older than 60 years. In older or immunosuppressed patients, HSV encephalitis can present non-specifically, with altered consciousness and without headache or fever. Other common forms of infectious encephalitis are associated with varicella zoster virus (VZV), West Nile virus (WNV), Japanese encephalitis virus (JEV), and enteroviruses, and can be differentiated based on local epidemiology and more distinctive individual phenotypes (figure 2B, 2C). For example, VZV encephalitis often presents with vasculopathy, leading to stroke; WNV can involve the anterior spinal cord, resulting in acute flaccid limb weakness; JEV is associated with a heterogeneous range of movement disorders; and enteroviral infections frequently involve the brainstem. Some forms of infectious encephalitis can present more chronically, often require a high index of suspicion, and typically reflect host vulnerability or delayed reactivation, rather than acute pathogen invasion.³¹ These subtypes are outside the scope of this Seminar.

In general, most forms of autoimmune encephalitis present less abruptly than infectious encephalitis, with combinations of cognitive and psychiatric features, seizures, movement disorders, and loss of awareness.^{15,19,32} Patient demographics and clinical profiles are often sufficient to distinguish between the common autoantigen-defined syndromes (figures 1, 3).^{15,19,20,33,34} In turn, this autoantigen-based classification informs the expected immunotherapy response, and the likelihood and nature of any underlying tumour (ie, paraneoplastic syndromes; figure 3B–C).^{35,36} Broadly, symptom onset over 3–7 days—with combinations of early and prominent psychiatric features that rapidly progress to seizures, cognitive dysfunction, or movement disorders—is most characteristic of N-methyl-D-aspartate receptor (NMDAR) antibody encephalitis, the most common form of autoimmune encephalitis in people younger than 40 years.³⁷ Myelin oligodendrocyte glycoprotein (MOG) antibody-associated encephalitis has a similarly acute presentation, occurs mostly in childhood, and is an increasingly recognised form of autoimmune encephalitis.^{38,39} Both of these subtypes can mimic infectious encephalitis as they often present with a viral prodrome of coryzal features, fever, and mild headache for a few days. By contrast, most other known autoimmune encephalitis subtypes present over weeks (or even months) without these systemic features. For example, leucine-rich glioma-inactivated 1 (LG11) antibody encephalitis—the most common subtype of autoimmune encephalitis in adults—typically affects people older than 40 years and usually presents over

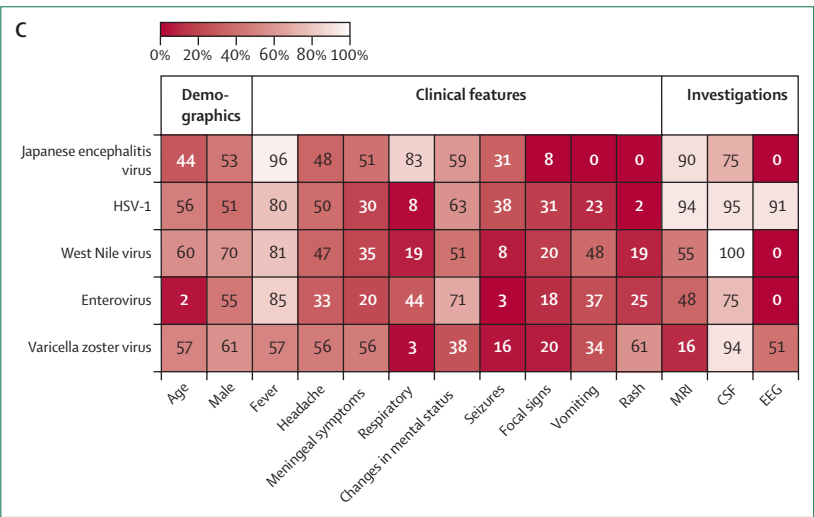


Figure 2: Acute and infectious encephalitis—features and approach
(A) Approach to diagnostic investigations in acute encephalitis, including safe expedited of CSF sampling^{102,108} and timely, parallel testing of both infectious and autoimmune aetiologies from blood and CSF. (B) Pathogen discovery timeline, describing vaccine-preventable (blue, above line) and currently non-preventable aetiologies (pink, below line). Black dots represent when the virus was isolated, grey dots show when the virus was discovered, and syringes indicate when the vaccine was discovered. Created with BioRender.com. (C) Key clinical features of the five most common infectious encephalitis viruses, with percentages of each feature displayed in each cell, extracted from the five largest studies of each virus. Median age of onset shown. Zeros are entered for unreported data. CSF=cerebrospinal fluid. EEE= Eastern equine encephalitis. EEG=electroencephalogram. GCS=Glasgow Coma Score. HSV=herpes simplex virus. JEV=Japanese encephalitis virus. VZV=varicella zoster virus. WNV=West Nile virus. *Mouse-brain derived. †Cell-culture derived.

several weeks with frequent focal seizures, amnesia, and disorientation.^{13,40–42} Similar features are present in contactin-associated protein-like 2 (CASPR2) antibody encephalitis, which predominantly affects males older than 60 years (figure 3).^{40,43} However, in women aged 18–40 years, several months of frequent focal seizures with subtle cognitive deficits is indicative of glutamic acid decarboxylase 65 (GAD65) antibody-associated encephalitis. Simple clinical observations can therefore aid diagnosis of autoantigen-defined encephalitis subtypes, which can help to direct and interpret first-line investigations (figures 1, 3, 4; appendix pp 17–29).

Clinical features in autoimmune encephalitis

The number of autoantibodies associated with autoimmune encephalitis has increased dramatically over the past two decades, and its current prevalence (ie, 10 cases per 100 000 population) is likely to rise further as new autoantibodies continue to be discovered (figure 3C).^{15,21,34,41,42} Many autoantibodies associated with autoimmune encephalitis are directed against intracellular targets (figure 3C; appendix 1 pp 21–24). These autoantibodies lack access to their targets in intact, live neurons and glia, and are therefore considered non-pathogenic.^{18,19} They represent valuable biomarkers of conditions likely mediated by T cells,⁴⁴ often indicating predictable clinical profiles and directing a search for specific remote tumours.^{35,36} For example, severe isolated cerebellar ataxia is often associated with Yo-antibodies

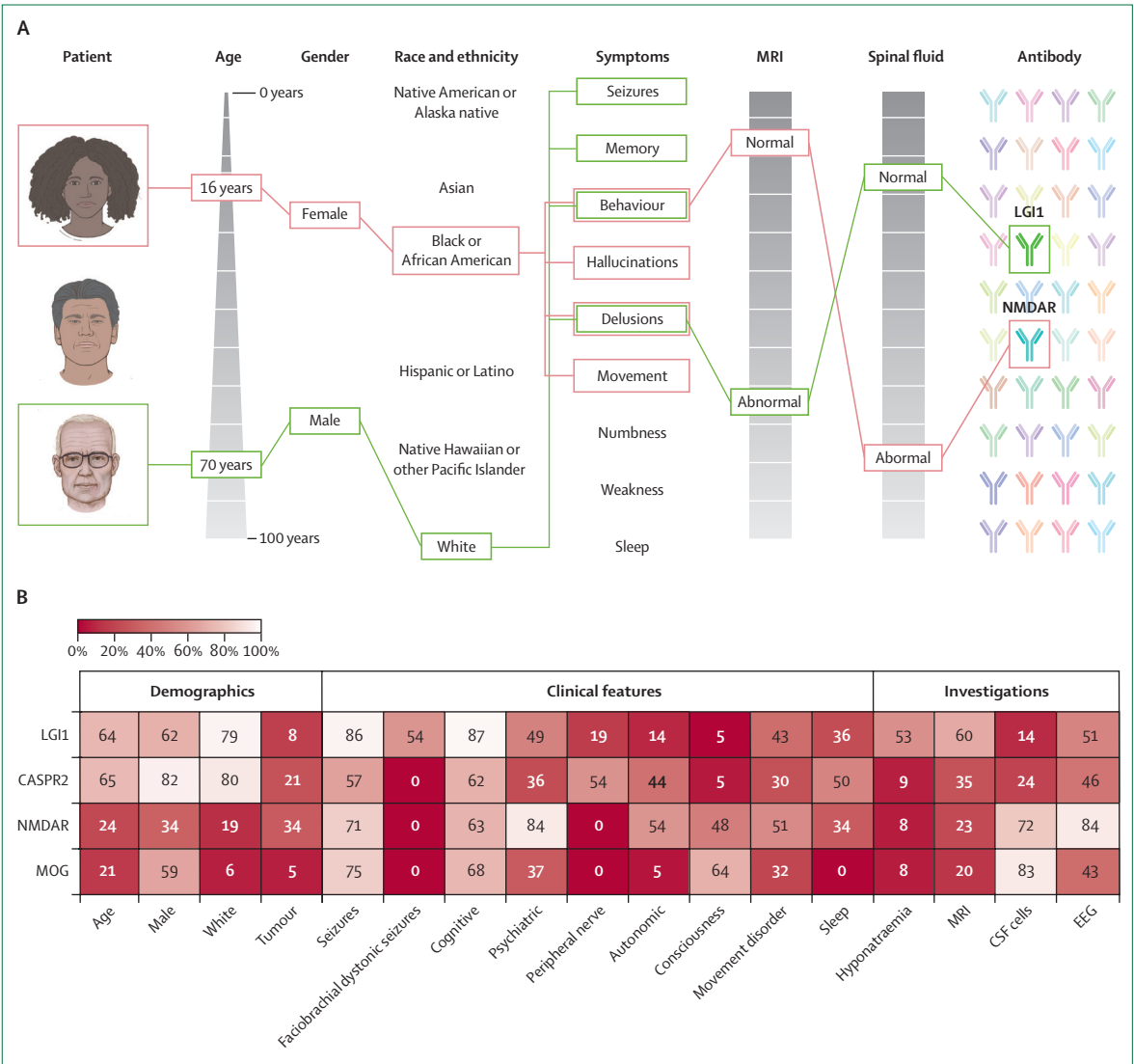
and uterine, fallopian, or breast cancer, whereas limbic-hypothalamic encephalitis with narcolepsy is associated with Ma2 antibodies and testicular cancer.³⁶

By contrast, autoantibodies directed against extracellular epitopes are likely directly pathogenic (figure 3C; appendix pp 17–20).^{15,18,20} The most common autoantigens (ie, LGI1, NMDAR, CASPR2, and MOG) each show distinctive features (figure 3B). As autoantibody results often take a few weeks to return, recognition of characteristic clinical features remains key to early diagnosis.

LGI1 antibody encephalitis is the most common form of adult-onset autoimmune encephalitis, with an incidence of approximately 2 cases per million people per year, a 2:1 male:female ratio, and a median age of onset of 64 years.^{13,40,42} Typically, LGI1 antibody encephalitis presents over a few weeks with increasingly frequent focal seizures,

often occurring several times per hour at disease nadir (ie, maximum disease severity; figure 4G). LGI1 antibodies are associated with a pathognomonic seizure appearance, termed faciobrachial dystonic seizures (FBDS), consisting of short-lived posturing episodes predominantly affecting the ipsilateral face and arm (figure 3B; video 1, video 2).^{13,45} FBDS are described by patients as jolts, jerks, or spasms, lasting 1–3 s, and can be dramatic, often associated with dropping items and falls. Typically, after a few weeks of increasing frequencies of FBDS, or other focal seizures, patients develop dense anterograde and retrograde amnesia with disorientation and behavioural changes. These symptoms can be mistaken for neurodegenerative conditions (eg, Alzheimer’s disease), particularly in the 50% of patients without medial temporal lobe swelling on MRI;⁴⁶ however, most dementias present over years with few seizures.¹⁶ Less than 5% of patients with LGI1 antibody

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(Figure 2 continues on next page)

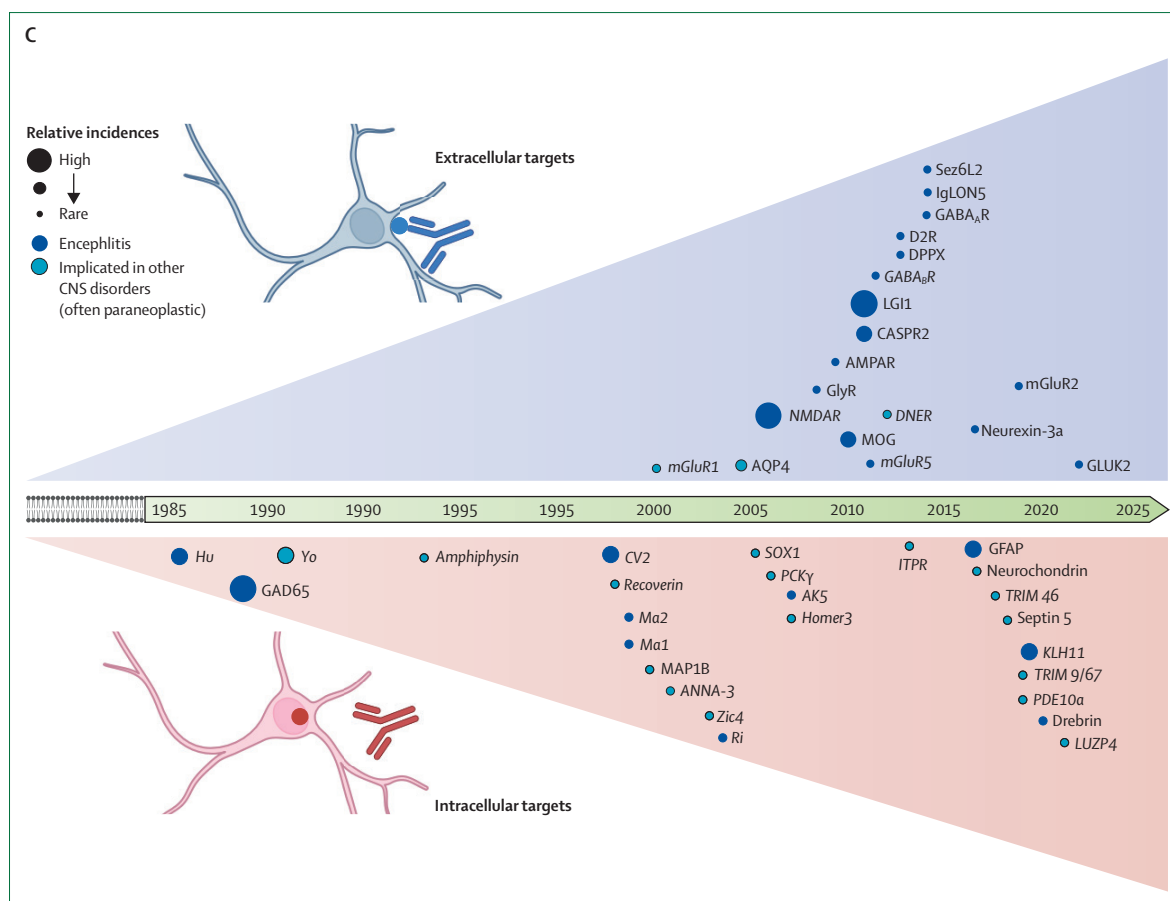


Figure 3: Autoimmune encephalitis features and discovery timeline

(A) Fundamental clinical features across demographics, symptoms, and simple investigations direct the detection of autoantigen-reactive antibodies. Examples shown with LGI1 and NMDAR antibodies, which are somewhat skewed towards White and non-White populations, respectively. Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved. (B) Demographics, clinical features, and investigations in autoimmune encephalitis. The percentages of each feature are extracted from the five largest studies of each of the five most common autoantibodies. Median age of onset shown. Zeros are entered for unreported data. (C) Timeline of discovery for autoantibodies targeting extracellular (above line) and intracellular (below line) epitopes. Many are implicated in autoimmune encephalitis. Created with BioRender.com. CSF=cerebrospinal fluid. EEG=electroencephalogram. GAD65=glutamic acid decarboxylase 65. LGI1=leucine-rich glioma-inactivated 1. MOG=myelin oligodendrocyte glycoprotein. NMDAR=N-methyl-D-aspartate receptor.

encephalitis have a tumour, and 50% have serum hyponatraemia, a potential early—albeit non-specific—diagnostic clue.¹³

CASPR2 antibody encephalitis has several similarities with LGI1 antibody encephalitis (figure 3B) as a late-onset, male-dominant disease (ie, an 8:1 male:female ratio), and both share strong yet distinct HLA-DRB1 allele associations.⁴⁷ However, patients with CASPR2 antibody encephalitis have fewer seizures than those with LGI1 antibody encephalitis, with more sleep disturbances, dysautonomia, and cerebellar involvement, and 50% show peripheral nerve involvement with hyperexcitability or neuropathic pain. For these patients, especially those diagnosed with Morvan's syndrome, a thymoma should be excluded.^{36,40,43}

NMDAR antibodies cause the most common form of autoimmune encephalitis in children and young adults,^{14,37,48} and show an approximate 30% association

with ovarian teratomas, partly accounting for the observed 3:1 female bias. The teratoma contains germinal centre-like structures and B cells that can secrete NMDAR antibodies, likely explaining the clinical benefits observed after early tumour removal.^{37,49} More rarely, NMDAR antibody encephalitis affects people older than 50 years, with a low rate of teratomas but a high rate of myriad malignancies, and a poor overall prognosis.^{37,50} Adults typically present over a few days with florid, multidomain psychiatric manifestations spanning affective, psychotic, behavioural, and sleep features, often resulting in early encounters with mental health services.^{14,37,48} However, within just a few days, almost all patients develop cognitive disturbances, seizures, or a complex movement disorder characterised by dystonia, chorea, stereotypies, and catatonia (video 3, video 4, video 5). This rapid onset and progression to neurological deficits typically distinguishes NMDAR antibody encephalitis from

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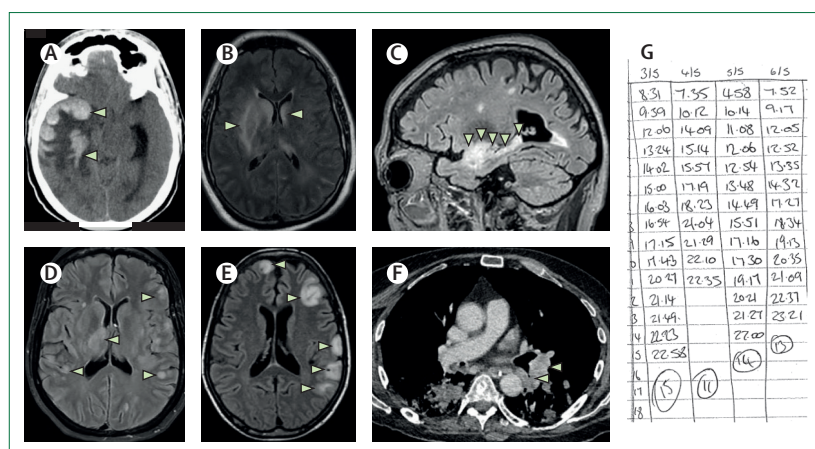


Figure 4: Brain imaging and patient diary in infectious and autoimmune encephalitis
 (A) CT head scan showing haemorrhage (arrows) and oedema (hypodense) in right temporal lobe secondary to herpes simplex virus-1 encephalitis. (B) Left and right basal ganglia involvement (arrows) with Japanese encephalitis virus. (C) Limbic encephalitis associated with autoantibodies against the GABA_A receptor, with high signal intensity in the hippocampus and amygdala on FLAIR imaging (arrows). (D) FLAIR imaging showing multifocal bilateral subcortical and right thalamic hyperintensities (arrows) associated with myelin oligodendrocyte glycoprotein antibodies. (E) Multifocal fluffy cortical and juxtacortical FLAIR hyperintensities (arrows) seen with GABA_A-receptor antibody encephalitis. (F) Left hilar lymph nodes (arrows) associated with small cell lung cancer and GABA_A receptor antibodies. (G) Diary of a patient with leucine-rich glioma-inactivated 1 antibody encephalitis recording 1 day of focal seizures per column. Daily totals are circled.

primary psychiatric diagnoses.^{51–54} Childhood presentations usually show less florid psychiatry and, rather, a rapidly evolving encephalopathy with more prominent seizures and movement disorders.^{55,56} After a few more days, both adults and children frequently require intensive care for depressed consciousness or central hypoventilation (or both), and approximately 30% of patients develop a multi-organ dysautonomia associated with potentially lethal cardiac dysrhythmias.^{14,37,48} Early screening for these typical sequelae, alongside appropriate monitoring measures, aims to reduce the consistently reported mortality rate of 5–10%.^{14,37,48,57} Despite the marked severity of this encephalitis, routine MRI is typically unremarkable.⁵⁸ Biomarkers of recovery in NMDAR antibody encephalitis, and other forms of autoimmune encephalitis, are still required to help titrate immunotherapy regimens.

In MOG antibody encephalitis—the next most common autoimmune encephalitis in children, which can also affect adults—patients show a short history of delirium, focal deficits, and seizures, often with a meningoencephalitis that can cause headache and neck stiffness.^{39,59} MRI is usually abnormal and can be highly characteristic, revealing cortical and subcortical swelling, with both parenchymal and meningeal enhancement (figure 4D).³⁹ Patients can also manifest synchronous or old optic neuritis and transverse myelitis, which are both well recognised associations of MOG-antibody associated disease. Tumours are rare, but the CSF will typically contain leukocytes with a high opening pressure.³⁸

GAD65 antibody-associated encephalitis predominantly affects young adult females with accompanying systemic

autoimmune comorbidities, who have a few months of progressive frequent focal seizures, with mild memory and affective disturbances but without prominent disorientation.^{41,60} This dominant phenotype of temporal lobe seizures and neuropsychiatric involvement without delirium is typically accompanied by unremarkable CSF and MRI findings. Serologically, very high concentrations of GAD65 antibodies are required to confirm neurological relevance, compared with the far lower concentrations found in people with type 1 diabetes and healthy individuals.^{60,61}

Other less common forms of autoimmune encephalitis are associated with antibodies against multiple neuronal and glial proteins, including γ -aminobutyric-acid (GABA)_A and β , glycine, and α -amino-3-hydroxy-5-methyl-4-isoxazole-propionic acid receptors (figure 3C; appendix pp 17–24).¹⁵ However, approximately 10–20% of patients with autoimmune encephalitis will lack a known autoantibody target; this subtype is known as seronegative autoimmune encephalitis.⁶² To avoid over-diagnosis of a syndrome with absent serological biomarkers, we encourage health-care practitioners to seek strict coherence between clinical, imaging, and CSF findings typical for autoimmune encephalitis.¹⁶ Patients with cancer who are treated with immune checkpoint inhibitors and CAR T cells can also develop autoimmune encephalitis as an immune-related adverse effect.^{63,64} For these individuals, management focuses on balancing the pressing need to treat the underlying cancer against risks of inducing neurological worsening or relapse.

Recent observations have temporally linked aetiologically distinct forms of autoimmune and infectious encephalitis with the recognition that acyclovir-treated HSV-1 encephalitis can precede—typically by 6–12 weeks—a form of autoimmune encephalitis most closely resembling NMDAR antibody encephalitis, and without detectable HSV-1 DNA in the CSF.^{65,66} This example of post-infectious autoimmunity occurs in approximately 25% of children and adolescents with HSV-1 encephalitis, with lower rates in adults, and is successfully treated with immunotherapy without further antimicrobials.

Clinical features in infectious encephalitis

Infectious encephalitis remains a substantial neurological burden worldwide, and is highly heterogeneous, varying by geography, season, and year.³ Five pathogens (ie, HSV, VZV, WNV, JEV, and enteroviruses) contribute a substantial proportion of cases globally, and their clinical (figure 2) and radiological features (appendix p 35) will be our focus herein. Previous reviews have comprehensively summarised more extensive ranges of infectious encephalitis.^{67,68}

JEV remains the leading cause of epidemic encephalitis in Asia, accounting for approximately 25000 deaths annually.⁶⁹ Although most human JEV infections are asymptomatic, particularly in vaccinated individuals, a minority develop clinically apparent disease with an acute

encephalitis.⁷⁰ Historically, this severe disease occurred predominantly in children; however, in regions with sustained paediatric vaccination programmes, adults are increasingly affected.⁷¹ The incubation period of JEV ranges from 5 to 15 days following an infective mosquito bite. Affected individuals typically experience a non-specific febrile prodrome of fever, headache, and vomiting, which can progress to delirium and seizures within several days.^{70,72–74} JEV shows marked neurotropism for deep grey matter structures, with lesions in the thalamus and basal ganglia frequently seen on both neuroimaging and neuropathology (figure 4B). Consequently, individuals with JEV often develop movement disorders, ranging from parkinsonism to hyperkinetic movements (eg, chorea and athetosis).^{74,75} These movements can resemble those seen in NMDAR antibody encephalitis (video 3, video 4, video 5), which can occasionally occur after JEV.^{24,76} Vaccination remains the most effective preventive measure for JEV, recommended by WHO for those living in or travelling to endemic areas, and can provide long-lasting protection.⁷⁷

HSV-1 is the leading cause of sporadic encephalitis in high-income countries, accounting for the majority of identified aetiologies.³ Unlike JEV, no effective HSV-1 vaccine exists, and long-term morbidity persists in approximately 70% of survivors despite acyclovir therapy.^{78,79} However, prompt intravenous acyclovir dramatically decreases mortality, making its empirical administration the standard of care. The optimal duration of acyclovir therapy remains uncertain; however, a minimum of 14 days of intravenous treatment is generally recommended, with longer courses (eg, 21 days) often used in severe cases or when clinical or virological concerns persist.^{79–81} Multiplex testing platforms can shorten the detection time for HSV-1, thereby reducing unnecessary empirical acyclovir use, improving cost-effectiveness, and decreasing the length of hospital stay.^{82,83}

WNV is the most common mosquito-borne cause of encephalitis in North America, with outbreaks also reported in parts of Europe, the Middle East, and Africa.⁸⁴ In the USA, neuroinvasive disease occurs with a 5% case fatality rate and 59% hospitalisation rate;⁸⁵ in Europe, 1340 locally acquired cases and 104 deaths were reported in 2022.⁸⁶ Approximately 80% of infections are asymptomatic. Although the majority of symptomatic cases are self-limiting febrile illnesses,⁸⁴ neuroinvasive disease occurs in less than 1% of individuals, primarily in adults older than 65 years and immunocompromised patients. These individuals present with fever, headache, vomiting, rash, altered mental status, and systemic laboratory abnormalities, including hyponatraemia, acute renal failure, and lymphopenia.^{84,87} Acute flaccid paralysis can be a characteristic feature of WNV, causing acute, asymmetric muscle weakness and subsequent wasting of the affected limbs. EEG, MRI, and CSF are frequently abnormal, with the white blood cell count in the CSF

typically exceeding 50 cells per μL .^{87,88} Neuroinvasive disease is confirmed by detection of WNV-specific IgM in the CSF. Currently, no vaccine or targeted antiviral therapy exists; management is supportive. Prevention relies on mosquito control, personal protective strategies, and surveillance of avian reservoirs to mitigate outbreak risks.

Enteroviruses, especially enterovirus-A71, are a major global cause of paediatric encephalitis, particularly in east Asia and southeast Asia.^{89–91} Outbreaks are seasonal in temperate climates but can occur year-round in tropical regions, with the geographical distribution varying by strain.⁹² Clinically, enterovirus encephalitis usually presents with fever, headache, altered mental status, and, in severe cases, coma. Specific strains are associated with distinct clinical phenotypes. For example, enterovirus-A71 has been linked to severe—and sometimes fatal—brainstem inflammation, whereas enterovirus-D68 has been implicated in acute flaccid paralysis.^{93,94} Diagnosis relies on the detection of CSF enteroviral RNA by PCR; serological testing is generally less sensitive. Because no specific therapies exist for enteroviruses, current management is primarily supportive, with ongoing research exploring antiviral and immunomodulatory strategies for people with severe infection.

VZV can cause both meningitis and encephalitis, with encephalitis occurring more frequently in older adults and immunocompromised individuals—demographics associated with poor outcomes.^{95–97} A vesicular rash can occasionally accompany VZV encephalitis.^{79,96} The condition commonly results in a vasculopathy affecting large or small vessels, leading to ischaemic or haemorrhagic strokes.⁹⁸ Neuroimaging can reveal frontal and temporal lobe oedema (which can mimic HSV-1 encephalitis; figure 4A),⁹⁹ and vascular imaging can show vessel wall irregularities or stenosis, consistent with a vasculopathy.¹⁰⁰ PCR of the CSF has suboptimal diagnostic sensitivity for VZV, whereas detection of VZV-specific IgG in the CSF is more sensitive; however, tests can be negative early in the disease course.^{101,102}

Diagnostic investigations

The four most important first-line investigations—blood tests, CSF studies, brain imaging, and EEG—show false positive and negative results in subtypes of autoimmune and infectious encephalitis; as such, they should be interpreted alongside well formed clinical impressions.

Other than the previously highlighted hyponatraemia associated with LGI1-antibodies, results of routine blood tests are typically normal in patients with autoimmune or infectious encephalitis, which helps to exclude systemic causes of encephalopathy, such as metabolic disturbances or sepsis (appendix p 2). An important exception is the presence of autoantibodies associated with autoimmune encephalitis, which, wherever possible, should be tested for in both the blood and CSF.⁴¹

CSF can provide direct insights into the brain; testing of CSF is indicated in all suspected cases of encephalitis, with the exception of cases that display the few contraindications to CSF sampling (eg, those with focal deficits, impaired consciousness, and signs of increased intracranial pressure; figure 2A). Spinal taps are usually well tolerated, only require local anaesthetic, take approximately 15 min with an experienced operator, and necessitate minimal post-procedure bedrest. Low-diameter blunt needles induce post-procedure headaches in fewer than 5% of patients.^{103,104} However, CSF is too often obtained late, principally due to concerns around headache and the rare complication of cerebral herniation. Simple recommendations indicate that rapid exclusion of absolute contraindications to CSF sampling (figure 2A) are proven to expedite the time taken to initiate antibiotic therapy, and hence reduce mortality and improve patient outcomes.^{105,106} Taken together, clinicians suspecting acute encephalitis should exclude the few features strongly suggestive of markedly raised intracranial pressure,^{107,108} urgently obtain CSF samples, and immediately thereafter administer intravenous acyclovir, plus consider empirical antibiotics to cover possible bacterial meningoencephalitis (figure 2A). In cases where CSF collection must be delayed until CT can be conducted, or in very agitated patients who might need sedation, acyclovir can be administered empirically.

An elevated CSF leukocyte count (ie, >5 cells/ μ L) often confirms the syndromic diagnosis of encephalitis. A normal CSF leukocyte count is useful to exclude infectious encephalitis; however, approximately 20% of patients with the condition can lack leukocytes, particularly those sampled early in their illness, or those who are older or immunocompromised.¹⁰⁹ To identify the specific cause of infectious encephalitis, CSF testing with both multiplex and targeted (monoplex) PCR assays is essential: multiplex panels enable rapid screening for a range of prespecified viral, bacterial, and fungal pathogens, whereas monoplex PCRs can provide higher sensitivity for specific pathogens when clinical suspicion remains high and panel results are negative. When unsuccessful, metagenomic next-generation sequencing can theoretically detect any DNA or RNA within the CSF, including from novel organisms.⁶⁸ However, false negatives can occur in patients with low nucleic acid burden in the CSF. Adjunctive serological, respiratory, or stool evaluations might still be required to identify some infectious pathogens, such as WNV, JEV, and eastern equine encephalitis.

Elevated CSF leukocyte counts by routine microscopy are only detected in some subtypes of autoimmune encephalitis (eg, NMDAR and MOG antibody encephalitis; figure 1). Flow cytometry is more sensitive and detects immune cells in many patients with acute autoimmune encephalitis; however, this technique is not widely available outside of research laboratories.^{110,111} CSF autoantibodies are a highly specific result in autoimmune

encephalitis,¹¹² but in some subtypes (eg, LGI1 antibody encephalitis), serum autoantibody testing is more sensitive.⁴¹ CSF protein and glucose concentrations are typically unremarkable in most forms of autoimmune and infectious encephalitis.

CT scans of the head are needed to facilitate CSF sampling in a minority of cases; however, their findings only guide diagnosis in very few forms of infectious encephalitis (ie, HSV-1 encephalitis-associated fronto-temporal intracranial haemorrhage and cerebral oedema; figure 4A) and very rarely in autoimmune encephalitis. Brain MRI is more useful than CT, as a normal MRI can usually exclude infectious encephalitis and abnormal patterns can substantially narrow down the specific causative pathogen (appendix p 35). Additionally, in several forms of autoimmune encephalitis, swelling predominantly affects the limbic regions (figure 4C), particularly the medial temporal lobes on T2-weighted fluid-attenuated inversion recovery sequences. When this swelling extends outside the hippocampal-amygdala regions, or is accompanied by diffusion restriction or contrast enhancement, infectious encephalitis is more likely than autoimmune encephalitis.⁴⁶ MRI can also show multifocal white matter lesions affecting subcortical regions and deep nuclei, such as those associated with MOG and GABA_A receptor antibodies (figure 4D, E). However, MRI results are unremarkable in more than 50% of common autoimmune encephalitis subtypes (eg, LGI1, CASPR2, and NMDAR antibody encephalitis; figure 1), which commonly prompts under-diagnosis.¹⁶ Upon suspicion of a paraneoplastic condition, CT or PET body imaging can aid the detection of a systemic tumour (figure 4F).

EEG is another valuable early investigation used to rule out treatable subclinical seizures and, in patients with autoimmune encephalitis who have normal MRI scans, to observe focal or diffuse slowing that support the diagnosis.^{32,113} EEGs are unremarkable in 20–60% of patients with autoimmune encephalitis, but abnormal in 50–90% of those with infectious encephalitis. Specific features, such as temporal lobe period lateralised epileptiform discharges in HSV-1 encephalitis, can be observed.^{113,114}

Management

A common theme in both autoimmune and infectious encephalitis is the time-dependent potential for recovery and the need to prioritise prompt administration of immunotherapies and antimicrobials. Supportive care can also involve strategic use of antipsychotics, mood stabilisers, benzodiazepines, and anti-seizure medications, to control the varied psychiatric, behavioural, sleep, movement, and seizure manifestations.¹¹⁵ In people with autoimmune encephalitis, the risk of an enduring tendency for seizures (ie, autoantibody-associated epilepsy) is generally less than 10%, typically mitigating the need for long-term anti-seizure medications.^{116,117}

However, epilepsy is more common after infectious encephalitis, and these patients often benefit from anti-seizure medications.¹¹⁸ Additionally, differential frequencies of tumours associated with subtypes of autoimmune encephalitis offer stratified approaches to prompt tumour identification and removal (figure 3).³⁶

Acyclovir (10 mg/kg three times daily for at least 14 days) remains the foundation for HSV-1 encephalitis treatment,¹² with the greatest benefit proven to occur when administered within 24–48 h of symptom onset.¹¹⁹ Acyclovir can safely be administered concurrently with dexamethasone (10 mg/kg four times daily for 4 days), albeit with minimal clinical benefits.¹²⁰ Other antiviral therapies indicated for specific pathogens include ganciclovir for cytomegalovirus encephalitis, and acyclovir or valacyclovir for VZV.⁵ For certain RNA viruses, including enteroviruses or flaviviruses, agents under evaluation include ribavirin and favipiravir.¹²¹

Immunotherapies are the mainstay of treatment for autoimmune encephalitis, and their early administration is consistently associated with improved outcomes across subtypes.^{13,14,122,123} The only completed randomised controlled trial confirmed a small, statistically significant benefit of intravenous immunoglobulins over placebo in both achieving seizure control and cognitive improvements in LGI1 and CASPR2 antibody encephalitis ($p=0.044$, odds ratio 10.5, 95% CI 1.1–98.9).¹²⁴ Therefore, although several trials are actively recruiting (appendix pp 30–34),¹²⁵ current evidence comes largely from retrospective observational data and expert consensus.^{115,122} Broadly speaking, this evidence has highlighted that each antigen-defined syndrome has relatively distinct

immunotherapy requirements. For example, most patients with LGI1 antibody disease show seizure cessation and improvements in both cognition and quality of life within a few weeks of first-line therapies, corticosteroids, plasma exchange, or intravenous IgG (or combinations thereof).^{13,40,126} However, in NMDAR antibody encephalitis, approximately 50% of patients show insufficient responses to these agents and benefit from the addition of second-line therapies, such as rituximab, cyclophosphamide, and ofatumumab, which reduce rates of disability, mortality, and relapse, and corticosteroid-induced side-effects.^{14,37,57,127,128} By contrast, benefits of immunotherapy appear minimal in GAD65 antibody encephalitis.⁶⁰ Across all autoimmune encephalitis subtypes, the optimal time to escalate or withdraw immunotherapies is not yet known.¹¹⁵

Multiple alternative immunotherapies for autoimmune encephalitis are either in development or in clinical trials (figure 5; appendix pp 30–34). Examples include monoclonal antibodies (eg, inebilizumab) or CAR T cells, which target CD19 (and are hence likely to delete broader B-cell lineages than anti-CD20 agents), CD38-directed daratumumab, which targets many B cells and plasma cells; proteasome inhibition (eg, bortezomib), which more selectively targets plasma cells compared with CD38-directed drugs; blockade of the pleiotropic IL-6 receptor with tocilizumab or satralizumab; and accelerated IgG degradation via FcRn blockade (eg, with efgartigimod).^{129,130} More precision experimental therapies include monovalent non-functional recombinant NMDAR antibodies, which block endogenous pathogenic autoantibody binding; direct

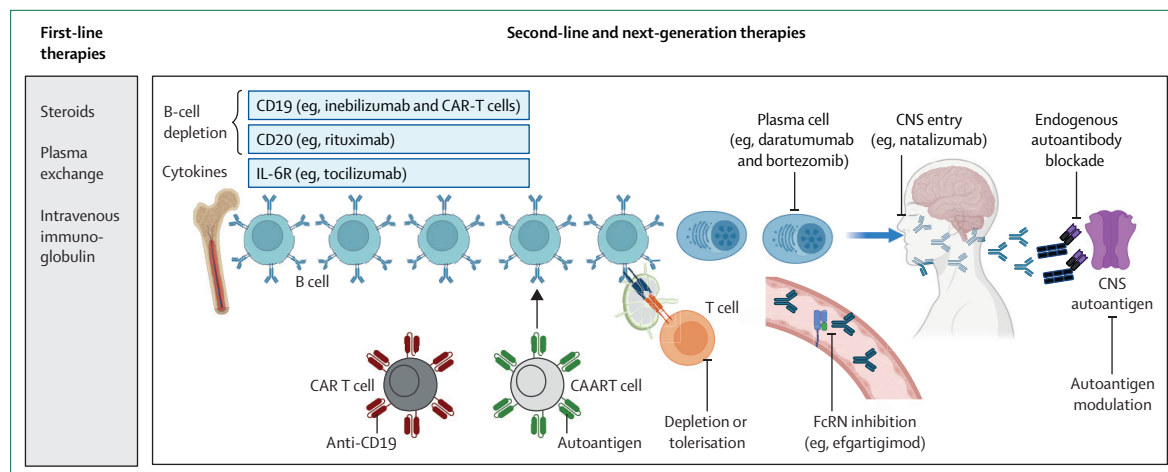


Figure 5: Established and emerging immunotherapies for treating autoimmune encephalitis

After first-line therapies, second-line immunotherapies should be considered, including B-cell depletion with CD19 or CD20-targeting medications, CAR T cells, IL-6R blockade with tocilizumab or satralizumab (ie, cytokine modulation), and drugs that delete plasma cells, including daratumumab (targeting CD38, which is also expressed on some B cells) and the proteasome inhibitor bortezomib. Emerging options include inhibition of leukocyte entry to the CNS (eg, natalizumab), blockade of FcRn IgG uptake to prolong the half-life of IgG (eg, with efgartigimod), blockade of endogenous antibody binding (eg, with monovalent antibody decoy therapy), direct pharmacological autoantigen binding to overcome autoantibody effects (eg, allosteric modulators), CAART cells (which express the autoantigen to induce selective deletion of autoantigen-reactive B cells), and various T-cell tolerisation or modulation approaches, to deny help to B cells. Depicted key neuroimmune compartments involved in the generation of pathogenic autoantibodies include bone marrow (the source of B cells and early T cells) and lymph nodes (a key site of crosstalk between B cells and T cells). CAR T=chimeric antigen receptor T cells. CAART=chimeric autoantibody receptor T cells. FcRn=Fc receptor neonatal. IL-6R=interleukin-6 receptor.

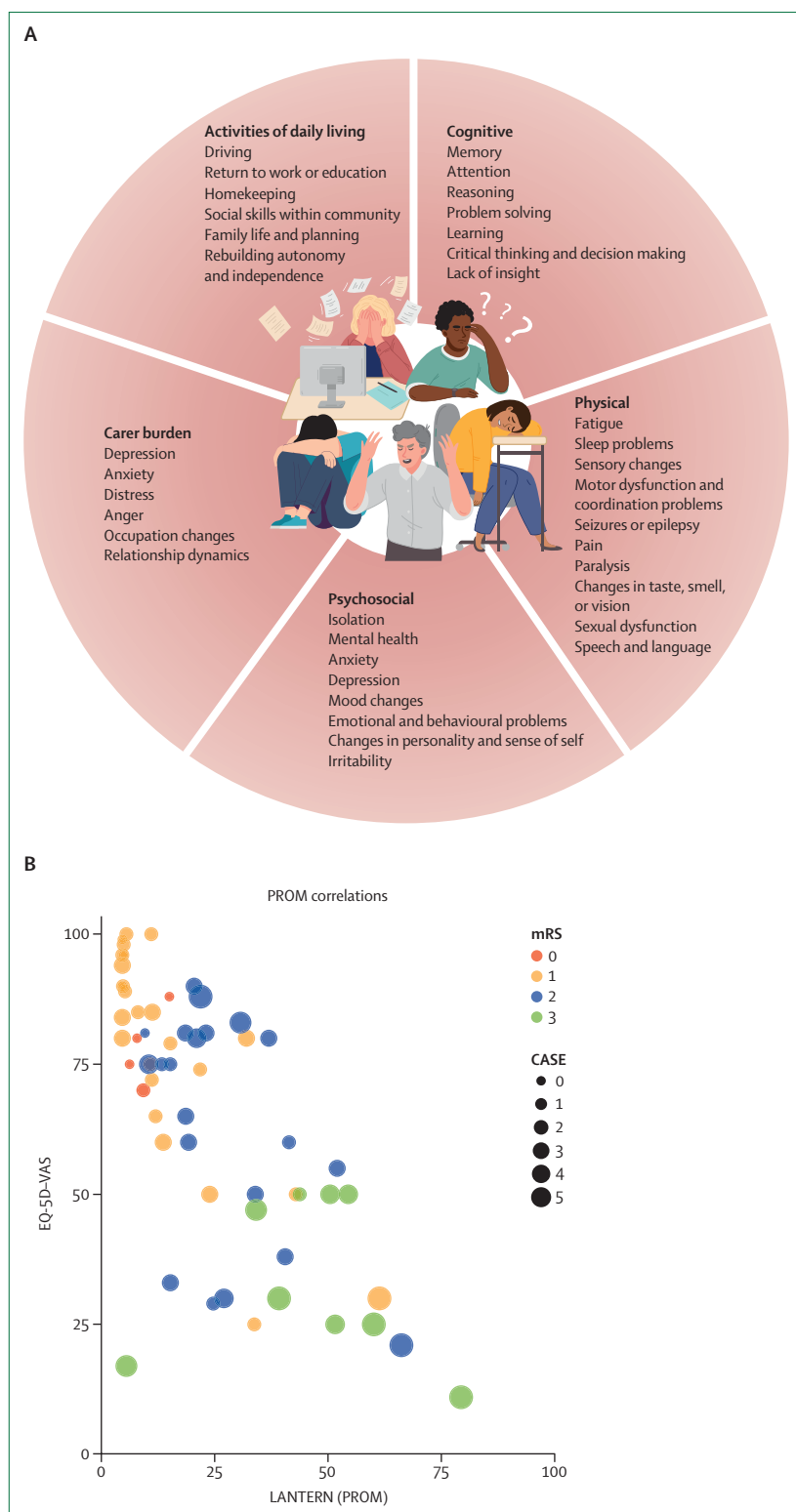


Figure 6: Patient outcomes after encephalitis
(A) Multiple functional domains are affected by infectious and autoimmune encephalitis. (B) In LGI1-antibody encephalitis, the patient-rated quality of life score (EQ-5D-VAS) correlates most closely with the patient-reported outcome measure (PROM), rather than the clinician-assessed modified Rankin score (mRS) or clinical assessment scale in autoimmune encephalitis (CASE).¹³⁵

modulators of autoantigen function;^{131,132} and chimeric autoantibody receptor T-cell therapies, which express the autoantigen and therefore aim to selectively deplete autoantigen-reactive B cells.¹³³

Outcomes

After acute (or, in some cases, long-term) immunotherapy treatment, the overall function of patients with autoimmune encephalitis improves for several years.^{13,14,43,134} For those with infectious encephalitis, improvements are more modest following treatment with antiviral agents, which is likely due to the less reversible nature of the original lesion. Regardless, most people with autoimmune and infectious encephalitis develop persistent—and often substantial—sequelae across cognitive, physical, psychosocial, and functional domains (figure 6A).^{135–137} Major residual neurocognitive problems involve memory, mood, personality changes, emotional regulation, attention, sleep, and fatigue, with varied relative preponderances across specific syndromes.^{134–140} Such deficits are likely under-recognised due to inadequate patient follow-up, encephalitis-induced communication and cognitive difficulties, and minimal exposure of patients to research, collectively downplaying the multifaceted and protracted experience of encephalitis survivors.

Relapses are a concern in autoimmune encephalitis, with rates of 10–50% depending on the subtype. Relapses can be challenging to identify, particularly in people re-presenting incomplete manifestations of the sentinel syndrome, such as seizures or psychiatric features alone. For these individuals, clinical decisions should be made alongside repeat investigations to distinguish between post-encephalitis manifestations and a genuine relapse, which would prompt immunotherapy to prevent deterioration.

In both adults and children, long-term functional consequences of encephalitis include failure to return to work or school, delayed education, and poor social reintegration, and are often only apparent after detailed neuropsychological testing, occupational health assessments, or caregiver interviews.^{136,138,141} The resultant delays in engaging with multiple relevant services can burden both patients and caregivers. Additional challenges include poor public awareness of encephalitis, under-recognised post-traumatic stress disorder after prolonged admission to intensive care units, and, for people with autoimmune encephalitis, pervasive anxiety around potential relapses.¹⁴² Despite these observations, comprehensive patient journeys remain poorly described in both autoimmune and infectious encephalitis, but can identify important features not captured by routine clinical assessments (see videos 7 and 8 in appendix p 3).

In autoimmune encephalitis, accurate outcome measurements are confounded by scores being imported from other disciplines.¹³⁶ The modified Rankin score (mRS), which is extensively reported in autoimmune encephalitis, was originally a clinician-assessed stroke

outcome tool used to capture physical and functional activities of daily living. The Clinical Assessment Scale for Autoimmune Encephalitis represents a more specific clinician-reporting tool and includes relevant domains such as seizures, conscious level, psychiatric features, and movement disorders, making it particularly well suited to NMDAR antibody encephalitis.¹⁴³ Only one dedicated patient-reported outcome measure has been reported in autoimmune encephalitis:¹³⁵ the LGI1-Antibody Encephalitis Rating scale correlates more closely with patient-rated quality-of-life than either clinician-rated or imported patient-rated scales (figure 6B). Global disability scales, such as the mRS, Glasgow Outcome Scale, Barthel Index, and generic health-related quality-of-life measures (eg, the EQ-5D), are commonly used in infectious encephalitis but are not validated in this setting.¹³⁷ To better incorporate cognitive and psychosocial sequelae, the Liverpool Outcome Score was developed specifically for paediatric encephalitis, and appears sensitive to detection of early recovery.¹⁴⁴ Patient-reported outcome measures specific to infectious encephalitis are awaited to enhance patient-centred clinical relevance.

Controversies and outstanding research questions

The past 25 years have witnessed major strides towards delineating causes of previously idiopathic forms of autoimmune and infectious encephalitis, establishing large cohorts to retrospectively analyse their management, and evaluate outcomes. However, multiple unresolved questions demand focused attention to guide future clinical and scientific directions and decrease the current substantial morbidity and mortality.

First, the aetiology of a large proportion of cases remains undetermined even with current molecular and serological techniques, highlighting the need for more sensitive and multiplex platforms for testing and diagnosing encephalitis, including low-cost point-of-care testing to better serve LMICs.^{145,146} Second, understanding of potentially modifiable host factors is inadequate, particularly for genetic and immunological susceptibilities in autoimmune and infectious encephalitis, despite these factors likely being paramount to understanding the pathogenesis of the conditions.^{47,147} Third, epidemiological data remain incomplete, especially in many LMICs, where under-diagnosis is prevalent given the insufficient contemporary diagnostic facilities and reporting systems (video 6).¹⁴⁸ Finally, rehabilitation and supportive interventions are under-explored, despite being beneficial for patients with encephalitis.

In infectious encephalitis, emerging and re-emerging aetiologies represent a major global health challenge, driven by viral evolution, environmental changes, and globalisation. Climate change has expanded the geographical range and activity of mosquito and tick vectors, increasing cases of WNV and JEV, and introducing Zika, Chikungunya, and Powassan viruses as causes of

infectious encephalitis in previously unaffected regions.^{149,150} Deforestation has increased human contact with wildlife reservoirs, facilitating spillover events such as high-mortality sporadic outbreaks secondary to Nipah virus, with high pandemic potential.¹⁵¹ Availability and uptake of vaccinations remains challenging, but vaccination programmes for pathogens such as JEV and tick-borne encephalitis markedly reduce the disease burden. Declines in vaccine coverage correspond to increased cases of measles encephalitis and subacute sclerosing panencephalitis, illustrating the severe consequences of vaccine hesitancy.¹⁵² Continued investment in vaccine deployment and vector control strategies are essential to mitigate these emerging threats. Reassuringly, vaccines have not been associated with new cases of neurological autoimmune diseases.^{153,154}

A diagnosis of autoimmune encephalitis is frequently conflated with several other clinical entities, including seronegative autoimmune encephalitis, Hashimoto's encephalopathy, paediatric autoimmune neuropsychiatric disorders associated with streptococcal infections, and subacute neurodevelopmental or regression syndromes.^{16,112,155–158} Additionally, isolated psychiatric features, brain fog, and subjective disorientation are highly unusual presentations of autoimmune encephalitis.^{15,53} Reports associating these syndromes and presentations with autoimmune encephalitis are usually a result of poor-quality autoantibody testing, including the use of non-native autoantigens, testing of CSF or serum in isolation, appreciable rates of low-specificity autoantibodies in healthy individuals and multiple disease control groups, a lack of objective inflammatory findings on CSF or MRI testing, and diagnostic constructs that are largely consensus-based and subject to change.^{16,53,112,159} Robust biomarkers and improved understanding of the pathobiology and disease boundaries of autoimmune encephalitis are therefore key areas for future study. Standardised autoantibody detection methods are also an important future aim; these methods should strive to retain conformationally native formats of autoantigens to mimic those recognised by IgG *in vivo*.^{16,20,34} In parallel, characterising the B-cell and T-cell biology underlying autoantibody production could improve understanding of the disease and guide novel diagnostics and therapeutics towards patient-centric care pathways.^{111,133,160,161}

Conclusion

The past few years have reclassified many enigmatic and idiopathic forms of encephalitis with molecularly precise, aetiology-based definitions. However, improved physician education regarding clinical presentations, targeted research agendas and funding to improve diagnostic capacities, and global public health initiatives are essential to meet the WHO-designated, urgent public health aim of reducing the overall encephalitis burden for survivors and their caregivers.

See Online for video 6

Contributors

SNMB, DS, AE, and KTT: conceptualisation, data curation, formal analysis, methodology, visualisation, writing (original draft), and writing (review and editing). SRI: conceptualisation, data curation, formal analysis, methodology, project administration, resource acquisition, supervision, visualisation, writing (original draft), and writing (review and editing). The authors confirm that the paper has not been submitted to another journal and has not been published in whole or part elsewhere.

Declaration of interests

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